Van-Thinh To

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EDUCATION

Erasmus Mundus Scholar - Double Degree Master's Program
University Paris Cité (France) *2025 - expected 2027*
Master of Bioinformatics **GPA: 18.05/20.00 (90%)** **First-class**
University of Milan (Italy) *2025 - expected 2027*
Master of Chemistry **GPA: 18.05/20.00 (90%)** **First-class**

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam) *2019 - 2024*
Bachelor of Pharmacy (5-year program) **GPA: 3.31/4.00 (83%)** **First-class**
Thesis: *A Knowledge-Guided Graph Self-Supervised Learning for Molecular Property Predictions* **Grade: 10/10**

RESEARCH EXPERIENCE

University Paris Cité (France)
Supervisor: Prof. Olivier Taboureau and Assistant Prof. Viet-Khoa Tran-Nguyen
Master's researcher student

1. **Large-scale PubChem/ChEMBL bioactivity curation and virtual screening benchmarking** *Sep 2025 - present*
   - Automating large-scale retrieval of bioactivity data for all protein targets from PubChem and ChEMBL.
   - Identifying dark chemical matter compounds for use as assumed inactives.
   - Conducting virtual screening benchmarking studies across selected protein targets.
2. **Target-specific machine learning scoring functions for cystathionine β-synthase inhibitors** *Jan 2026*
   - Curated data set from PubChem and ChEMBL.

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)
Supervisor: Assoc. Prof. Tuyen Ngoc Truong and PhD. Researcher Tieu-Long Phan
Undergraduate researcher

1. **Generative AI for dual targets of EGFR and VEGFR** *May 2024 - August 2025*
   - Developed a knowledge-guided graph-based scoring function to evaluate generated molecules.
2. **A knowledge-guided graph for molecular representation** *Dec 2023 - Jul 2024*
   - Introduced pre-text tasks for self-supervised learning guided by orbital information.
   - Proposed a novel knowledge-guided graph (KGG) utilizing a hierarchical molecular graph, defragmentation, and two knowledge vectors that incorporate orbital information for molecular representation.
   - A 2.4% improvement in ROC-AUC for classification tasks, along with a 39%, and 14% reduction in RMSE, and MAE respectively for regression tasks, compared to a state-of-the-art model on the MoleculeNet dataset.
3. **A novel application of machine learning in virtual screening of HIV integrase inhibitors** *Jun 2022 - Jan 2024*
   - Identified the applicability domain of the machine learning models using Principal Component Analysis for dimension reduction, coupled with convex hull techniques.
4. **Application of molecular similarity, neural network and GNINA molecular docking in virtual screening of PD-L1 inhibitors** *Jun 2023 - Dec 2023*
   - Proposed an artificial neural network model, and a molecular similarity assessment; utilized GNINA 1.0 molecular docking.
   - Developed an automated pipeline for the molecular similarity model.
   - Screened 15,235 DrugBank repurposing compounds and identified 22 hit compounds.
5. **Machine learning and Deep neural networks with consensus molecular docking for enhanced potency prediction of ALK inhibitors** *Dec 2022 - Jun 2023*
   - Trained an ensemble model including a graph neural network, an artificial neural network and a XGBoost model for the prediction of potential Anaplastic Lymphoma Kinase inhibitors.

SELECTED PUBLICATIONS

Full publication list

1. **To, V.-T.**, Van Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. F. & Truong, T. N. KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions. *Journal of Chemical Information and Modeling.* doi:`10.1021/acs.jcim.5c01068` (2025)
2. Van Nguyen, P.-C., **To, V.-T.**, Tran, N.-V. N., Phan, T.-L., Truong, T. N., Gärtner, T., Merkle, D. & Stadler, P. F. SynCat: molecule-level attention graph neural network for precise reaction classification. *Digital Discovery.* doi:`10.1039/D5DD00367A` (2026)
3. **To, V.-T.**, Phan, T.-L., Doan, B.-V. N., Nguyen, P.-C. V., Le, Q.-H. N., Nguyen, H.-H., Trinh, T.-C. & Truong, T. N. Innovative virtual screening of PD-L1 inhibitors: the synergy of molecular similarity, neural networks and GNINA docking. *Future Medicinal Chemistry,* 1–12. doi:`10.1080/17568919.2024.2389773` (2024)
4. Phan, T.-M., Phan, T.-L., Van-Nguyen, P.-C., Le, H. S. L., **To, V.-T.**, Truong, T. N., Merkle, D. & Stadler, P. F. ProQSAR: A Modular and Reproducible Framework for Small-Data QSAR Modeling with Fit-and-Use Models. *Journal of Cheminformatics.* doi:`10.1186/s13321-026-01175-9` (2026)
5. Phan, T.-L., Trinh, T.-C., **To, V.-T.**, Pham, T.-A., Van Nguyen, P.-C., Phan, T.-M. & Truong, T. N. Novel machine learning approach toward classification model of HIV-1 integrase inhibitors. *RSC advances* **14,** 14506–14513. doi:`10.1039/D4RA02231A` (2024)
6. Truong, G.-B., Pham, T.-A., **To, V.-T.**, Le, H.-S. L., Nguyen, V. P.-C., Trinh, T.-C., Phan, T.-L. & Truong, T. N. Discovery of VEGFR-2 Inhibitors employing Junction Tree Variational Encoder with Local Latent Space Bayesian Optimization and Gradient Ascent Exploration. *ACS Omega.* doi:`10.1021/acsomega.4c07689` (2024)
7. Trinh, T.-C., Phan, T.-L., **To, V.-T.**, Pham, T.-A., Truong, G.-B., Le, L. H. S., Tran, X.-T. D. & Truong, T. N. Synergy of advanced machine learning and deep neural networks with consensus molecular docking for virtual screening of anaplastic lymphoma kinase inhibitors. *Journal of Computer-Aided Molecular Design* **39,** 79. doi:`10.1007/s10822-025-00657-6` (2025)

TEACHING EXPERIENCE

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam) *Aug 2024 - Aug 2025*

Contracted teaching assistant and research mentoring - Department of Organic Chemistry, Faculty of Pharmacy

- Wrote research proposals for four university-funded projects, each awarded 1200 USD.
- Prepared training materials on computer-aided drug design for undergraduate students, such as `RDKit`, fingerprints, molecular descriptors, ANN, GNN.
- Mentored one undergraduate student on thesis project focused on self-supervised learning.
- Taught practical organic chemistry, with a focus on laboratory equipment proficiency, ChemDraw usage, lab techniques and synthesis skills.

SKILLS

- **Cheminformatics:** `RDKit`; Machine Learning (`Scikit-learn`); Deep learning (`TensorFlow`, `PyTorch`); Molecular Docking (`GNINA`); Generative diffusion model (`IRDiff`); Property-matched decoy generation (`DeepCoy`); Databases (PDB, PubChem, ChEMBL); Visualization (`PyMOL`)
- **Programming skills:** Programming language (Python, R, C), Linux terminal, Pytest, Git, Package management, deployment
- **IELTS:** Overall 7.0 (Academic Version) *Nov 2024*

TRAINING

Vienna Summer School on Drug Design (Austria) *Sep 2023*

- Gained foundational training in drug discovery using docking, pharmacophore and molecular simulation tools (LigandScout, KNIME, OpenEye)
- Learned concepts of techniques in drug discovery, such as generative models.

TALKS

1. **To, V.-T.**, Van-Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. & Truong, T. *KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions* in *41st Scientific and Technical Conference of Pharmacy* (2025)(Oral - Presenter)
2. **To, V.-T.**, Phan, T.-L. & Ngoc Doan, B.-V. *Application of Molecular Similarity and Artificial Neural Networks for PD-L1 inhibitors Virtual Screening* in *ECMC2023: 9th International Electronic Conference on Medicinal Chemistry* (2023) (Poster - Presenter)

SELECTED AWARDS

- **Pharmacy Scholarship for Elite Students** *2019 - 2024*
  Awarded a $5000 stipend over 5 years, covering full tuition (acceptance rate: 10%)
- **Second prize in the nationwide Vietnamese Chemistry Olympiad** *2018, 2019*
  The biggest Chemistry competition for high school students in Vietnam with 400+ participants, ranking 42 overall
- **Odon Vallet Scholarship** *2019, 2024*
  Awarded to outstanding high school, university, and graduate students in Vietnam.

REFERENCES

Associate. Prof. Tuyen Ngoc Truong
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PhD. Researcher Tieu-Long Phan
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